Firas Mostafa

AI & Backend Developer | Python · Django · LLM & RAG Systems

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SUMMARY

Software Engineer (2026 graduate, AI specialization) building AI-driven backend systems with Python, Django, and PyTorch. Experienced with RAG pipelines, LLM integration (LangChain + Ollama), gradient boosting ensembles, real-time WebSocket APIs, vector search (pgvector), and Docker. Designed and built a multi-container health AI platform solo as a graduation thesis, from model training to load-tested API endpoints. Available for remote and on-site positions.

TECHNICAL SKILLS

- **Backend & APIs:** Python, Django REST Framework, Daphne (ASGI), FastAPI, Celery, Redis, WebSockets, JWT, TDD
- **AI & Machine Learning:** PyTorch, XGBoost, LightGBM, CatBoost, RAG (LangChain + Ollama + BGE-M3), Faster-Whisper ASR
- **Databases & Infrastructure:** PostgreSQL + pgvector, Redis, Docker, Docker Compose, Git, GitHub Projects (Agile/Kanban)
- **Mobile & Frontend:** Flutter, Dart, Flutter Web, BLoC / Cubit, GetX, Provider, Google Maps & Google Fit SDKs, Firebase

EXPERIENCE

Freelance Flutter Developer — Food Delivery Platform (Client Project) | *Flutter, GetX, Google Maps, Flutter Web*

Sep 2024 – Jan 2025 • Sole developer

- Customized a white-label multi-vendor food delivery platform (StackFood: customer, restaurant and delivery apps, built with Flutter and GetX) for a client, working within a large, undocumented third-party codebase.
- Got the apps running on the web with fully responsive layouts, upgraded the Flutter SDK and dependencies, fixed build and runtime issues, and deployed the web apps to Hostinger.

PROJECTS

Ursa — AI Health Assistant & Triage Platform (Graduation Thesis) | *Django ASGI, PyTorch, LangChain, Docker, Flutter*

Jan 2026 – Sep 2026 • Solo project • Demo video: youtu.be/WBz_ei6zKjU

- Architected a multi-container backend (Docker Compose): Django + Daphne ASGI gateway, Celery workers with Redis broker, PostgreSQL + pgvector, an Ollama LLM container, and Faster-Whisper ASR.
- Built a medical RAG pipeline: BGE-M3 embeddings → pgvector search over physician-verified Altibbi (Arabic) & MedQuAD (English) corpora → prompt augmented with patient vitals and safety guardrails → LLM response streamed over WebSockets.
- Trained a ResNet-1024 MLP ensemble (5-seed averaging + residual connections) on 246K Kaggle records across 721 disease classes (Top-1: 85.97%, Top-3: 95.88%). Built a calibrated tri-model severity classifier (XGBoost + LightGBM + CatBoost with soft-voting) on 558K Kaggle emergency triage records (ROC-AUC: 0.9295, Critical Recall: 94.00%).
- Engineered 15 physiological features (Shock Index, MAP, Pulse Pressure, Hypoxia Index) and implemented a dual-voice OR-Gate safety protocol combining ML ensemble predictions with LLM clinical risk assessment.
- Designed an FSM for triage conversation flow, an intent router for Arabic/English queries (8.1K cleaned Arabic training samples), and Celery Beat follow-up workflows with automatic escalation.
- Wrote 80 unit and integration tests (100% pass rate). Load: 87.72 req/s sustained, 525 ms avg, 0% errors. Spike: 100 concurrent users, 707 ms avg response.

TasteCraft AI — AI-Powered Recipe & Cooking Assistant | *Django REST, JWT, Docker, PostgreSQL, Flutter*

May 2025 – Sep 2025 • github.com/firas-mostafa/tastecraft-ai

- Engineered a Django REST backend with JWT authentication, Docker containerization, AI-driven recipe generation, and comprehensive TDD test coverage. Built the companion cross-platform Flutter app.

AgriSphere — Smart Agriculture & Crop Health Management | *Flutter, REST APIs, ML Integration*

Sep 2025 – Dec 2025 • Team project • github.com/firas-mostafa/agri_sphere

- Integrated REST APIs and ML model outputs into a Flutter app to deliver real-time crop health diagnostics.

EDUCATION

Al-Sham Private University — Damascus, Syria

B.Eng. in Informatics Engineering (AI Specialization) | 2021 – 2026

LANGUAGES

Arabic: Native • **English:** Professional working proficiency (strong written)